Kavya Pratap Singh

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Education

Noida Institute of Engineering & Technology (NIET) — **Greater Noida, India**
Bachelor of Technology in Computer Science & Engineering ***(Current CGPA: 8.67 / 10)*** — *2024 – 2028 (Ongoing)*

- Core Subjects: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks.
- Leadership: **Vice President – Design Thinking Club** (Organizing student tech bootcamps & hackathons).

Senior Secondary School (CBSE) — **India**
Class XII – Science (Physics, Chemistry, Mathematics, Computer Science) ***(Score: 85%)*** — *2023 – 2024*

Secondary School (CBSE) — **India**
Class X – High School ***(Score: 95%)*** — *2021 – 2022*

Technical Skills

Languages: Python, Kotlin, JavaScript, TypeScript, Dart, Java, SQL, C/C++, HTML/CSS, LaTeX
AI & Machine Learning: Vertex AI (Google Cloud), Gemini Multimodal Vision API, Local LLMs (CodeLlama, Qwen-Coder via Ollama/GGUF), Deep Learning (CNN & ANN), Prompt Engineering, OCR
Mobile & Web Development: Android SDK, Jetpack Compose, Room Database (Offline-First), Firebase (Firestore, Auth), Next.js, React.js, Node.js, RESTful APIs, Jinja2, Tailwind CSS
Developer Tools & Cloud: Google Cloud Platform (ADC), Google Play Console (Verified Developer Account), Git, GitHub, Linux/Bash, Docker, Vercel
Spoken Languages: English (Professional Working), Hindi (Native/Bilingual), German (Elementary / Currently Learning)

Featured Projects

Quesper – AI Exam Paper & Question Bank App | *Android, Kotlin, Jetpack Compose, Room, Firebase* **Nov 2025 – Present**

- Privately engineered and launched a native Android platform enabling school teachers to assemble print-ready CBSE/NCERT exam papers in under **3 minutes** (saving 2+ hours per test) by tapping questions directly from authentic pages.
- Architected an **offline-first local cache** with Room Database paired with automated cloud sync via Firebase Firestore.
- Built an automated blueprint calculation engine supporting custom section limits, Bloom's taxonomy marks weighting, and instant export to print-ready PDF and editable Word (.docx) formats.
- Published and actively maintained on Google Play via verified **Google Developer Account**; Live: quesper.in.

Quesper AI – Universal Book Extractor & Question Synthesizer | *Vertex AI, Python, Gemini Vision, LaTeX* **July 2026**

- Privately developed an autonomous multimodal AI pipeline using **Vertex AI (Gemini 3.7 & 3.5 Flash via gcloud ADC)** that extracted **20,000+ authentic questions** directly from 300 DPI textbook pages with zero-solution filtering.
- Synthesized CBSE/NCERT curriculum-aligned **Case-Based Questions, MCQs, and Assertion-Reason Questions** with step-by-step verified solutions.
- Implemented an automated bounding-box diagram segmentation engine (4.5% padding) and a battle-tested mathematical **LaTeX formula corrector** repairing syntax corruptions and backslashes.

Tkinter App Designer – Visual GUI Studio & Local AI Synthesizer | *Python, Tkinter, Local LLMs, Jinja2, Flutter* **2024**

- Privately built a desktop GUI designer in 2024 with an interactive drag-and-drop canvas, snap-to-grid alignment, marquee multi-widget selection, undo/redo history stack, and an instant **F5 Live Preview** runner.
- Integrated an **offline Local LLM code synthesizer** (CodeLlama / Qwen-Coder) for private, zero-latency callback function generation and automated API deprecation repairs.
- Built an extensible multi-target compiler utilizing Jinja2 templates, outputting clean, production-grade code for **Python Tkinter/CustomTkinter, Flutter (Dart)**, and **Java (Swing)**.

Experience

Proprietary Software Systems & EdTech Platforms — **Noida / Remote**
Full-Stack & AI Systems Developer (Privately Built Applications) — *2024 – Present*

- Privately developed proprietary desktop developer tooling, multimodal AI vision pipelines, and native Android software.
- Built Quesper AI (July 2026) using Google Cloud Vertex AI, extracting **20,000+ curriculum questions** with high-precision bounding box cropping and LaTeX mathematical formula repairs.
- Developed Quesper Android (Nov 2025 – Present) with Room DB offline-first architecture and Firebase cloud synchronization, published under verified Google Play Developer account.

Achievements & Leadership

Vice President – Design Thinking Club (NIET): Leading innovation workshops, creative design bootcamps, and hackathons.
2nd Position – Calculus of Innovation Hackathon: Awarded 2nd position for architecting and pitching a cutting-edge AI EdTech solution.
NPTEL Certifications: Completed rigorous academic technical certifications with elite scores through IIT / Ministry of Education.
Google Developer Account: Verified Google Play Console developer account publishing and maintaining production Android software.